HiMCM 2018 Problem B — Summary Sheet

Team GoHiMCM AeroTherm-MPC sources: `bang_vs_mpc_summary.json`, `harness_benchmark_results.csv`, `agent_thermostat_dialogue.json`

Background. TOU electricity tariff peaks punish reactive cooling. Bang-Bang hysteresis short-cycles the compressor. We build AeroTherm-MPC on a 2R2C thermal network. Wall mass stores pre-cooling before peak rates.

Model and algorithm. Envelope capacitances are $C_{\text{air}} = 6.2 \times 10^5$ and $C_{\text{wall}} = 1.8 \times 10^7\,\text{J/°C}$. COP is $3.8 - 0.04(T_{\text{out}} - 25)$. Comfort uses ISO 7730 PMV with penalty outside $|\text{PMV}| > 0.5$. TOU rates are \$0.12/\$0.22/\$0.48 (peak 14:00–19:00). Zero-order hold at $T_s = 15\,\text{min}$ yields prediction matrices. Model predictive control (MPC) uses horizon $H = 24$. SciPy L-BFGS-B solves each window. Cognitive agents may retune the comfort band via Ollama or Mock.

Teaching-day results (locked).

Controller	Energy	Cost	Peak kWh	Peak share	Starts
Bang-Bang	6.232	\$1.682	1.514	43.2%	42
AeroTherm-MPC	**6.223**	**\$1.611**	**1.274**	**38.0%**	**3**

Cost −4.2%. On-peak energy −15.8%. Starts 42 → 3. Prepeak $\bar{u} = 0.40$ vs peak $\bar{u} = 0.17$.

Harness (scenarios A–E). Mean M_1 cost is \$2.427 (Bang) vs \$2.340 (MPC). Mean M_3 mechanical score is 77.6 vs 6.4. Heatwave starts fall from 130 to 7. Sensor-noise starts fall from 133 to 16.

Cognitive MAS. Consensus band is $[21.5, 25.0]$°C with λ scale 1.25. MAS-tuned MPC cost is \$1.603 (−4.67% vs Bang). Starts fall to 2. Source in the locked file is Mock fallback.

Impact. Predictive pre-cooling shifts cooling off the TOU spike. Homeowners can raise the afternoon setpoint by about 1.5°C. See the homeowner advisory letter for three household rules.